FLIGHT 1.6

AUTOPILOT GUIDE

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1 Introduction

This document is a guide of how to set up, enable and use the horizontal steering autopilot in Flight version 1.6. It covers use with EIVA's ROTVs: the ScanFish and the ViperFish.

The autopilot is used for minimizing survey time by actively steering the ROTV horizontally towards a runline using the 3D flaps. By importing a cross-track error (ROTV position relative to runline) and setting up simple parameters for the runline tracking, the operator only needs to engage and disengage the autopilot before and after each runline and the flap control happens automatically based on commonly used autopilot algorithms.

Below you will learn how to import the cross-track error, how to configure the autopilot, and get an overview of the interface and usage of the autopilot.

2 Autopilot importer in ScanFish Flight

The autopilot importer in Flight is where the cross-track error (XTE) is set up. It is also where the host network IP and port and string prefix are configured.

The horizontal-steering autopilot works based on the distance off line (DOL) data being sent as an NMEA XTE sentence from EIVA NaviSuite.

The DOL data contains the location of an object's reference point relative to a survey line, typically the USBL beacon position in the centre of the ROTV. The distance can be either in metres and nautical miles. When the ROTV is in operation, and the USBL beacon updates the navigation software with an updated position – typically every one-to-two seconds – that DOL will change. The updated data is then sent from the navigation software to Flight software, and the autopilot will then adjust the physical flap angle of the horizontal steering flap motor on the ROTV, based on the configuration of the autopilot in Flight.

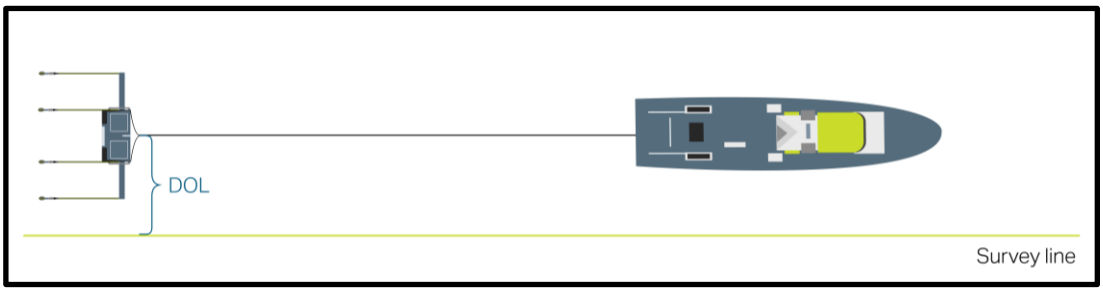

Figure 1 Distance off line illustration

Within the NaviSuite software suite, NaviPac outputs the XTE message for the Flight importer.

In the NaviPac configuration, set up a new data output driver by selecting the plus (**+**) on the ROTV instruments window. Select one of the 5 NMEA slots available on the ROTV object as shown in Figure 2 below.

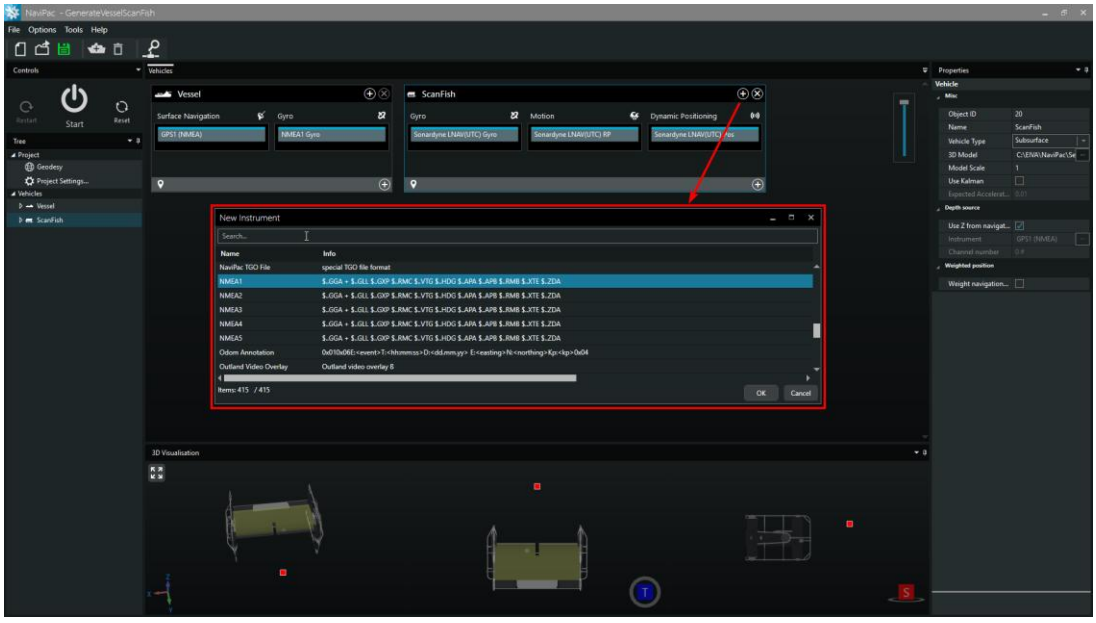

Figure 2 Configuring an NMEA XTE output in NaviPac

Next, the I/O needs to be setup with the driver selected. The default port reserved for the importer in Flight is port *9110*, and if Flight is running on the same computer as NaviPac, the localhost (*127.0.0.1*) can be set as the IP, see Figure 3.

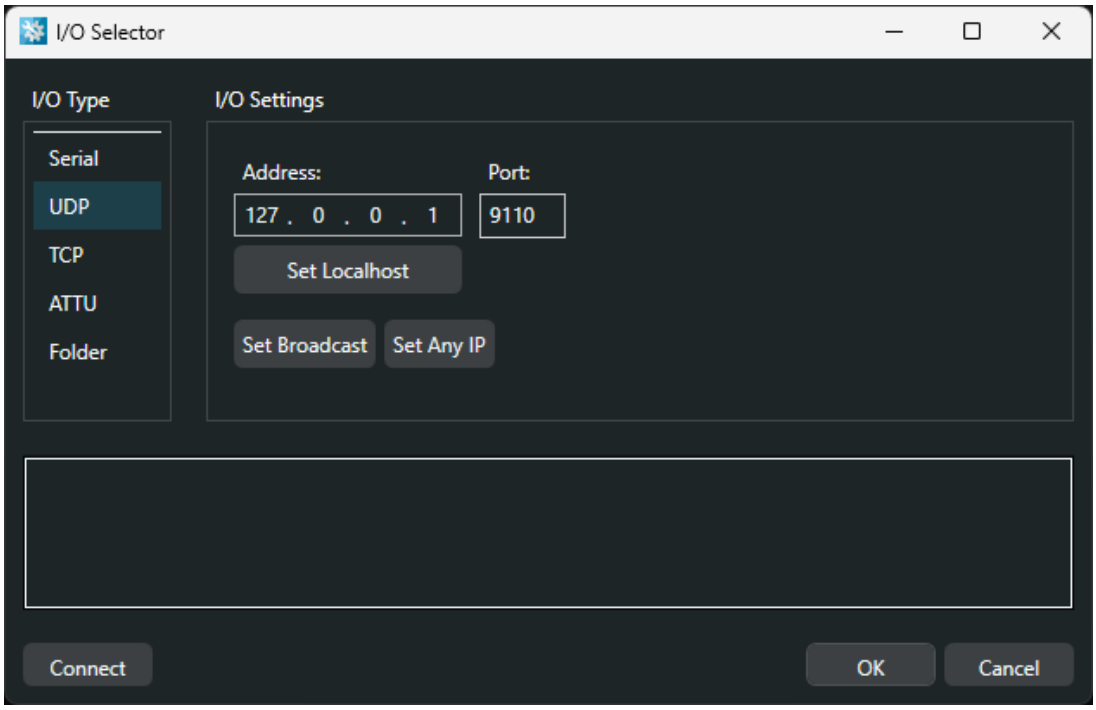

Figure 3 I/O Selector for the NMEA output driver

Under the **Properties** of the driver, there are several areas to ensure are correctly assigned:

- **Source**: *CRP* or a different POI
- **NMEA Telegrams** > **String prefix**: *NP* or *GP* are both accepted
- **Autopilot**: XTE selected, and GGA unselected as this is default

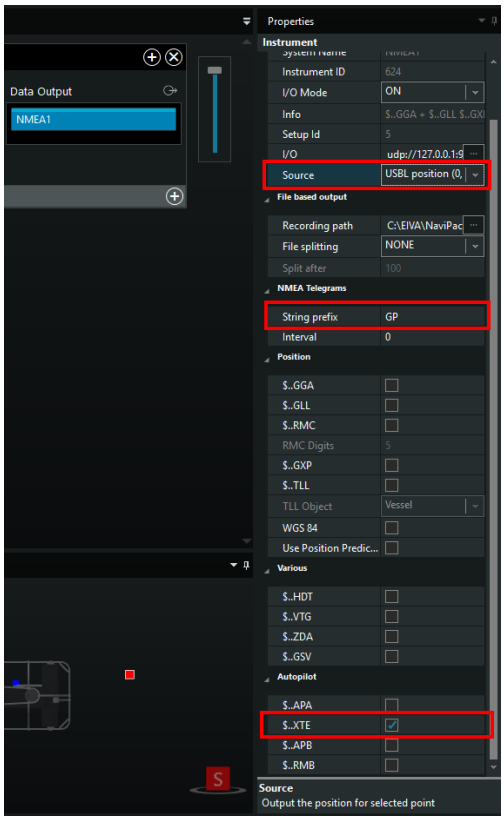

Figure 4 Selecting the right properties for the output driver in NaviPac

Go online in NaviPac and select a runline under the ROTV in **Vehicle Control**, and the NMEA XTE sentence will now be sent over port 9110.

Next, you need to set up the importer in ScanFish Flight.

In Flight, you need to set the importer up under **Settings** > **Options**… > **Importers** > **Autopilot importer** > **Import ports,** shown in Figure 5.

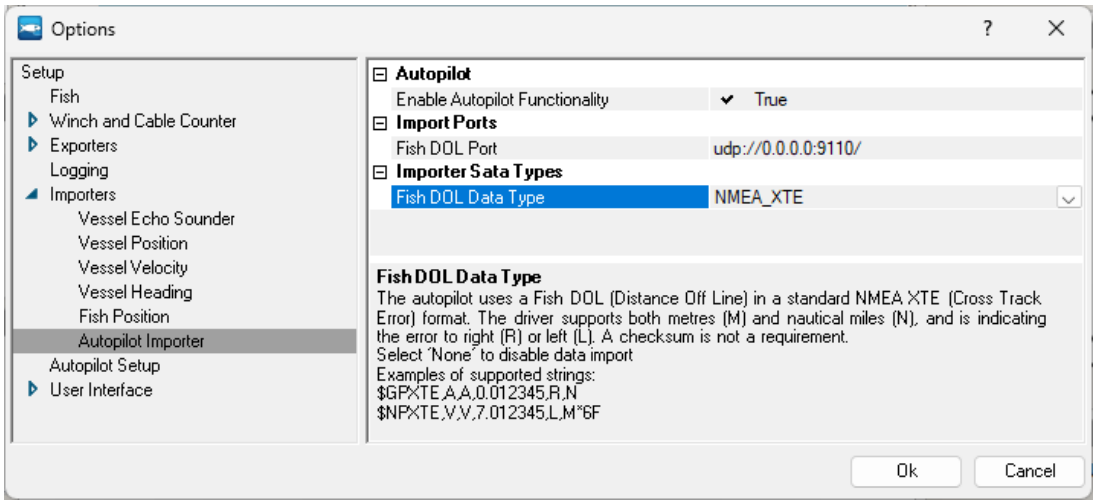

Figure 5 Setting the autopilot importer up in ScanFish Flight

The measured cross track error is a part of the RMB message, for compatibility with some older equipment designed to work with Loran. The format of the XTE string, as examples show in the image above, are as follows:

$GPXTE,A,A,3.4,L,N,W*hh<CR><LF>

$GPXTE,1,2,3,4,5,6<CR><LF>

1. Status:
    - A = Valid
    - V = Loran-C Blink, SNR warning, general warning flag
2. Status:
    - A = Valid
    - V = Loran-C Cycle Lock warning flag
3. Cross track error magnitude
4. Direction to steer:
    - L = Left
    - R = Right
5. Cross track units:
    - N = Nautical mile
    - M = Meters
6. FAA mode indicator – Optional (NMEA 2.3 and later)

For more information, please refer to the NMEA 0183 standard.

Inside the IO Selector, you can test the data is being sent and received correctly by selecting **Test**.

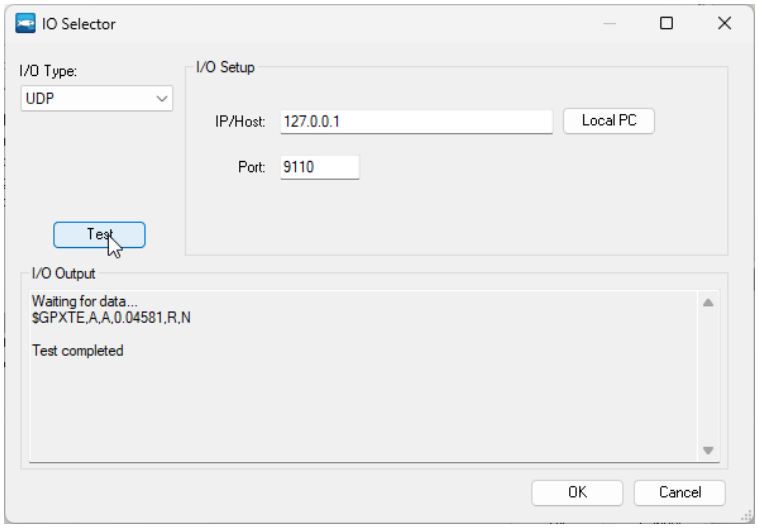

Figure 6 IO Selector with Test selected

The **IO Selector** in Flight can be opened via **Options** > **Import Ports** > **…**.

3 Configuring the autopilot

The settings for the horizontal autopilot steering are configured in the **Autopilot Setup** menu.

Open **Autopilot Setup** via **Settings** > **Options** > **Autopilot Setup**..

Misc	
Algorithm Used For The Autopilot	Segment + PID
Segment Regulation Settings	
Default Segment Angle	0 degrees
Angle Limit	40 degrees
Off Distance Count Limit	10
Adjusment Window Min	2
Adjusment Window Max	15
Angle Adjustment Value	2 degrees
Distance Offset	1 meters
Angle Multiplier	2
Angle Offset	0 degrees
Segment Display Count	5 Segments
Quick Angle Offset	5 degrees
PID Regulation Settings	
Proportional Gain	0.4
Integral Gain	0.3
Derivative Gain	0.1
Integral Limit	1
DOL Offset	
Enable DOL Offset	✔ True
Quick DOL Offset	10 meters

Figure 7 Configurable default settings for the ROTV autopilot

The first step in configuration is choosing which algorithm to use for the autopilot.

Under **Misc**, you can choose the algorithm via **Algorithm Used For The Autopilot** option. There are two algorithms to choose from: **Segment regulation** and **PID**.

The autopilot options can be either turned off, each algorithm enabled individually, or both enabled.

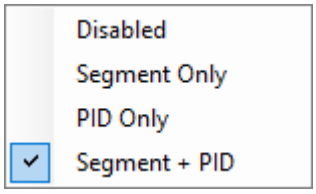

Figure 8 Selecting algorithm for the autopilot

The table below describes the remaining settings in the **Autopilot Setup**.

Segment Regulation Settings	A segment is defined as the distance between two points of the cross-axis of the survey line, so called bins. Each segment or bin has a flap angle associated with it, so when the ROTV is in a segment, its 3D flaps are set to the angle associated with the segment. See Figure 9 below.
Default Segment Angle	Defines the angle the segment control uses at startup for segment 0 (survey line). It is recommended to keep this at zero.
Angle Limit	Defines the ± maximum rudder angle that the autopilot regulation can use. **Note**: The flap motors have a physical angle limit around 40 degrees, so setting a value higher than this will have no effect. Setting conservative angle limits may be necessary in cases where external equipment has a high surface area, resulting in roll effects to the system when the 3D function is engaged.
Off Distance Count Limit	Defines how many samples are inside the **Adjustment window** before an adjustment should happen. A lower number will give a more reactive autopilot.
Adjustment Window Min	This parameter defines the minimum number of meters from the target survey line the **Off distance count limit** should increase by. Set this to a ± value that you are satisfied the ScanFish should stay inside without making changes to the **Segment regulation** algorithm. **Note**: It is not recommended to have a zero here.
Adjustment Window Max	This parameter defines the maximum number of meters from the target survey line, the **Off distance count limit** should increase. This parameter is a way to reject spikes from the USBL positioning system.
Angle Adjustment Value	Defines how many degrees the flap angle value should change when an adjustment happens. A higher number will give a more aggressive autopilot, but it soon becomes too aggressive.
Distance Offset	Bin size in metres of each segment.
Angle Multiplier	Multiplier used to calculate the resulting flap angle. $Flap\ angle = Angle\ multiplier * (DOL + Angle\ offset)$
Angle Offset	A constant used in above equation.
Segment Display Count	Number of segments that should be displayed
PID Regulation Settings	A Proportional-Integral-Derivative regulator is a feedback-based control loop mechanism commonly used to manage machines and processes that require continuous control and automatic adjustment.
Proportional Gain	The proportional gain constant dictates how much the system should react to the latest incoming DOL data. The proportional gain constant produces an output value that is proportional to the current distance to the target value. The proportional response is adjusted by multiplying the error by this constant.

	A proportional gain value of **1** will result in a 1:1 ratio of the distance from target in meters to the flap angle. Increasing the value to **2** will result in a 1:2 ratio, meaning if the distance from target is 2 meters the result sent to the ScanFish will be **4**. Setting the proportional gain too high will result in a system that oscillates around the survey line.
Integral Gain	The contribution from the integral term is proportional to both the magnitude of the error and the duration of the error. The integral is the sum of the instantaneous error over time and gives the accumulated offset that should have been corrected previously. The accumulated error is multiplied by the integral gain constant and added to the output value. It serves as a memory buffer for the past values sent to the flaps and helps accelerate the movement towards the desired position. An integral gain factor that is set too high may result in the ROTV overshooting the survey line, as the accumulated error values are larger than the proportional value.
Derivative Gain	The derivative error is used to predict the systems behaviour, helping the system settle on the correct survey line and overall system stability.
Integral Limit	Sets a maximum limit to the integral gain to avoid it running out of control.
DOL Offset	Allows the ROTV to have an offset from the survey line. Ideal for dual ScanFish setups or gap filling.
Enable DOL Offset	Enables the quick DOL offset buttons and entry for manual typing offset value.
Quick DOL Offset	Defines the quick offset button's value to starboard and portside (coloured green/red) if enabled.

Table 1 Autopilot settings explained

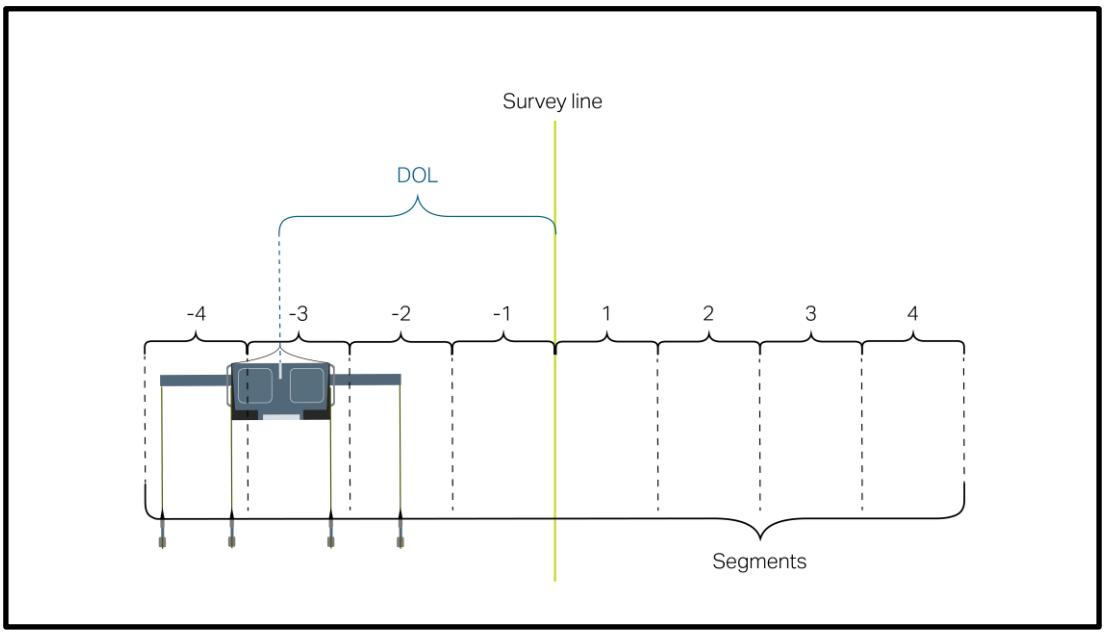

Figure 9 Illustration of segments in the autopilot

When the autopilot is configured, the ScanFish **Flight toolbar** will display a button that toggles between **Engage Autopilot** and **Disengage Autopilot**.

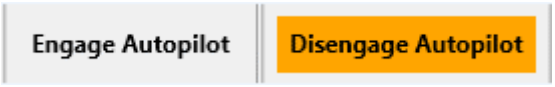

Figure 10 Engage/Disengage Autopilot button on the ScanFish Flight toolbar

4 Autopilot view

The autopilot overview panel in ScanFish Flight is where the autopilot information is displayed, such as cross track error information, flap angle value and if the autopilot is engaged. Simple configurations like resetting the algorithms and DOL offset is configured from here.

To open the **AutoPilotPanel** view go to **View** > **Docable Panels** > **AutoPilotPanel**

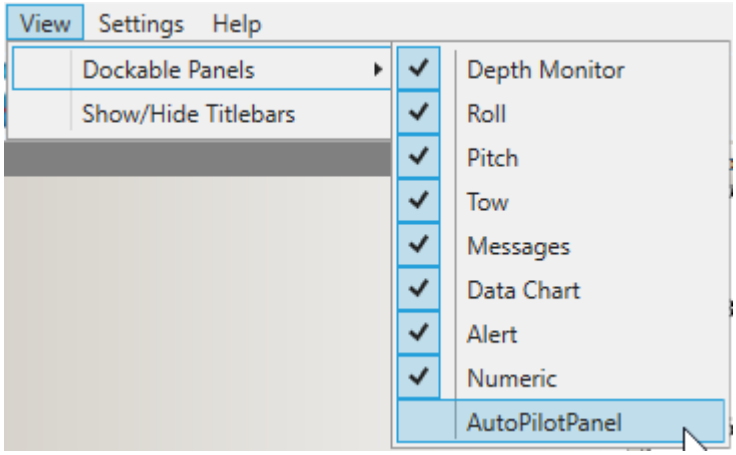

Figure 11 How to open the Autopilot view

If it is connected in Flight and the XTE is being sent to Flight, the **AutoPilotPanel** should now show the Distance from line in metres, as highlighted below.

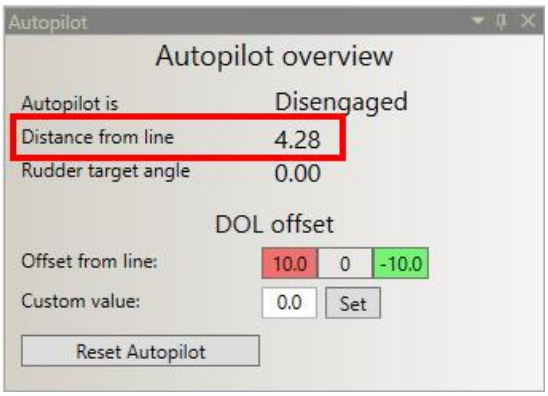

Figure 12 Autopilot window in Flight

5 Autopilot usage

The **Engage** / **Disengage Autopilot** button is part of the **ScanFish Flight toolbar**, and is displayed with an orange background when engaged to indicate the autopilot is activated.

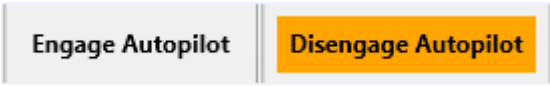

Figure 13 Engage/Disengage Autopilot button

The **AutoPilotPanel** shows the status of the ROTV horizontal steering autopilot. The information displayed changes when engaged / disengaged from the toolbar.

Figure 14 shows the disengaged state of the **AutoPilotPanel**. This will only display the **Autopilot overview** and **DOL offset** sections (if enabled in settings).

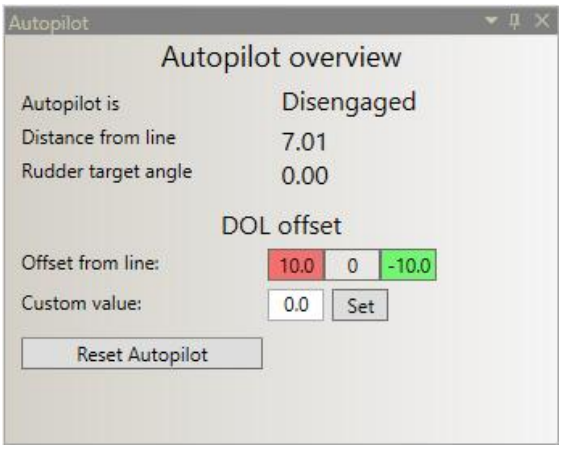

Figure 14 Autopilot panel when disengaged

Figure 15 shows the engaged state of the **AutoPilotPanel**. Now the **Segment regulation** and **Parallel PID regulation** sections are displayed if configured for both algorithms. Each algorithm's contribution to the autopilot is displayed in each of their sections.

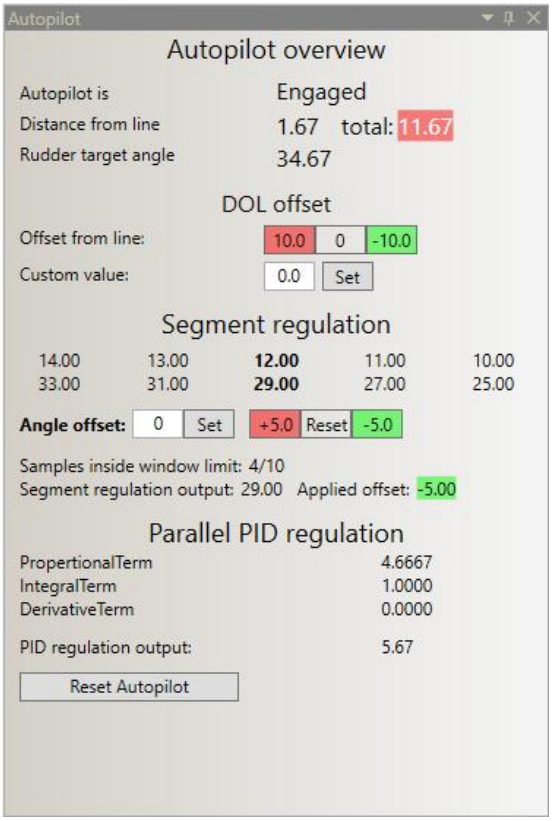

Figure 15 Autopilot panel when engaged

Table 2 below explains each section of the **AutoPilotPanel** view.

Autopilot overview:	
Autopilot is (*Engaged*/*Disengaged*)	Displays if the autopilot is activated or not.
Distance from line	Displays the DOL in metres from the NMEA XTE interfaced for the autopilot.
Rudder target angle	Displays the total rudder target-angle output from the autopilot algorithm(s).
DOL offset	
Offset from line	Quick-buttons (values are user-configurable) for steering the ROTV to an offset relative to the survey line defined in the NMEA XTE sentence. This is convenient in dual ScanFish setups, where the same survey line is used for both systems, but offsets to each side for greater coverage.
Custom value	As an alternative to user-configurable quick **Offset from line** buttons, the user may type in a custom value in this field. Click **Set** to apply the value. **Caution**: be aware that the value that is typed in affects the outcome of the cross-track error message (XTE) and does not specify the intended offset. This means a negative value will bring the ROTV the specified number of metres to starboard of the

	line. A positive offset will force the autopilot to aim for the specified number in metres to port of the runline the XTE sentence is aiming for. Colours (green/red) clarifying the resulting side of the runline the autopilot is aiming for in the **Autopilot overview** section **total**.
Segment regulation	
Segment display count	In the top row, different segments are displayed. In the bottom row, the resulting flap angle for each of the segments are displayed. The bold central column shows the ROTV's current segment and resulting rudder angle from the **Segment Control** algorithm.
Angle offset	The user can force the ScanFish faster to one side by either manually typing in an angle offset and clicking **Set,** or by using the quick buttons that add the rudder angle value to the **Segment Control** algorithm. When an angle offset is applied, the total offset is shown under **Applied offset**.
Parallel PID regulation	
Proportional Term	The proportional gain contribution to the PID regulation.
Integral Term	The integral gain contribution to the PID regulation.
Derivative Term	The derivative gain contribution to the PID regulation.
PID regulation output	The total PID output based on all three inputs.
Reset Autopilot	The **Reset Autopilot** button will reset the autopilot to start from neutral when reengaged. If not reset, the autopilot will pick up from the last PID and/or segment input. **Note**: It is recommended to reset the autopilot before starting a new survey line.

Table 2 Autopilot view sections explained